

Ark Routes

SatsPath treats Ark as a public payment rail and keeps execution separate from route planning.

Direct Ark Receive

An Ark receiver can publish an Ark public pointer:

- Ark server URL,
- compressed receiver public key,
- optional VTXO pointer,
- optional ownership proof.

If the sender is compatible with the same Ark server, the route planner can produce an **ArkToArk** intent. This does not require Boltz, Lightning, or an on-chain swap. Current SatsPath CLI output is `preview/intents` unless explicitly testnet-executed and supported by the local bridge.

Ark Swap Routes

Ark swaps are different from direct Ark receive:

- **ArkToLightning** requires an offboard/submarine-style path.
- **LightningToArk** requires an onboard/reverse path.
- **ArkToOnchain** requires Ark offboard support.
- **OnchainToArk** requires Ark onboard support.

SatsPath does not call a normal Bitcoin address an Ark receive. If the Ark bridge cannot create or claim the required Ark-side object, the route is intent-only and fails closed for execution.

Current Status

- Ark route planning exists.
- Ark direct receive/send previews exist.
- Ownership proof verification exists.
- Mainnet execution is disabled.
- Testnet execution is gated by `--testnet`, `--execute-testnet`, exact confirmation text, profile validation, Ark ownership proof validation, bridge availability, encrypted swap storage, and claim/refund builder availability.
- Boltz Ark swap settlement is not implemented end to end.